

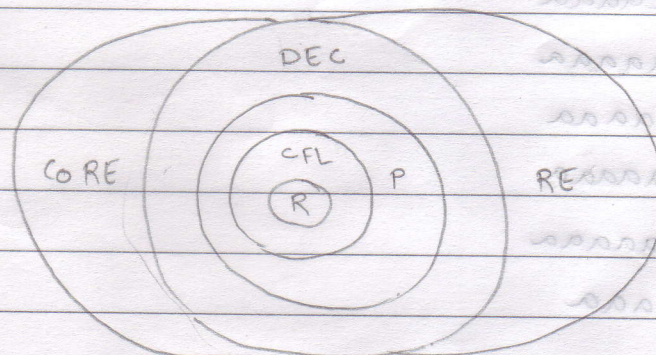
FOUNDATION OF COMPUTING THEORY - HOMEWORK 08.

Q1)

Closed under	Complement	Concatenation	Intersection
P	YES	YES	YES
CPL	NO	YES	NO
RE	NO	YES	YES
Decidable	YES	YES	YES
Regular	YES	YES	YES

Q.2)

- a) The decidable languages = DEC
- d) Language whose complement is Turing-recognizable = CORE
- e) complement is decidable = DEC (since it is also decidable)
- g) Turing-recognizable languages = RE
- f) Regular languages = R.



Q.3.

$$S \rightarrow ACaB$$

$$\Rightarrow AaaCB$$

$$\Rightarrow AaaDB$$

$$\Rightarrow AaDaB$$

$$\Rightarrow ADaaB$$

$$\Rightarrow ACaaB$$

$$\Rightarrow AaaCaB$$

$$\Rightarrow AaaaaCB$$

$$\Rightarrow AaaaaDB$$

$$\Rightarrow AaaaDaB$$

$$\Rightarrow AaaDaB$$

$$\Rightarrow AaDaaB$$

$$\Rightarrow ADaaaaB$$

$$\Rightarrow ACaaaaB$$

$$\Rightarrow AaaCaB$$

$$\Rightarrow AaaaaCaB$$

$$\Rightarrow AaaaaaaCaB$$

$$\Rightarrow AaaaaaaaaCB$$

$$\Rightarrow AaaaaaaaaE$$

$$\Rightarrow AaaaaaaaaEa$$

$$\Rightarrow AaaaaaaaaEaa$$

$$\Rightarrow AaaaaaEaaa$$

$$\Rightarrow AaaaaEaaaa$$

$$\Rightarrow AaaaEaaaaa$$

$$\Rightarrow AaaEaaaaaa$$

$$\Rightarrow AaEaaaaaaaa$$

$$\Rightarrow AEaaaaaaaaa$$

$$\Rightarrow aaaaaaaaa$$

Q4) Assuming $EQ_{CFG, DFA}$ is decidable.

~~Then there exists a TM R which decides EQ_{CFG} .~~

Let R be the TM that accepts $EQ_{CFG, DFA}$

Constructing an algorithm for a TM S that uses R to decide ALL_{CFG} .

S = "On input $\langle G \rangle$, where G is a CFG

(i) Construct a DFA D that accepts language $= \Sigma^*$

(ii) Run R on $\langle G, D \rangle$ ~~where~~

(iii) If R accepts, accept; if R rejects, reject.

$\therefore$ If R decides $EQ_{CFG, DFA}$, S decides ALL_{CFG} .

But ALL_{CFG} is undecidable.

$\therefore$ By contradiction.

$EQ_{CFG, DFA}$ is undecidable